

Table 1. *G36-1wk* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Medium-sized office building	Chicago, IL, USA (ASHRAE climate zone 5A)	Simulated	G36	359 (127, 105, and 127 faults in cooling, heating, and shoulder season, respectively.)	114	1-min	1 week

Table 2. Details of *G36-1wk* fault scenarios in the cooling and shoulder seasons. Faults were injected continuously for a total of one week, but the system operated in occupied mode only from 7 a.m. to 8 p.m. each day.

Category	Fault abbreviation (CSV file name)	Fault type	Fault intensity	Method of fault imposition
Sensor	TCHWSup_p1	Chilled water supply temperature sensor bias	+1 °K	Sensor faults were decomposed into three parts: multiplicative errors, bias, and noise, as defined below. $V_{EP} = (1 + m) \cdot V_{EF} + \varepsilon_{bias} + \varepsilon_{noi}$ where V_{EP} and V_{EF} represent the error-presented and the error-free values, respectively. m is the multiplicative offset of the scaling error. ε_{bias} and ε_{noi} denote the deviation caused by the bias errors and the noise.
	TCHWSup_m1		-1 °K	
	TCHWSup_p2		+2 °K	
	TCHWSup_m2		-2 °K	
	TCWSup_p1	Cooling water supply temperature sensor bias	+1 °K	
	TCWSup_m1		-1 °K	
	TCWSup_p2		+2 °K	
	TCWSup_m2		-2 °K	
	RelPreCHW_p5000	Chilled water differential pressure sensor bias	+5,000 Pa	
	RelPreCHW_m5000		-5,000 Pa	
	RelPreCHW_p10000		+10,000 Pa	
	RelPreCHW_m10000		-10,000 Pa	
	Vout_p15	Outdoor air flow rate sensor scale error	+15%	The output of the original sensor model was overwritten by the faulty value (V_EP) when the fault mode was activated.
	Vout_m15		-15%	
	Vout_p30		+30%	
	Vout_m30		-30%	
	TSup_p1	AHU supply air temperature sensor bias	+1 °K	Both positive and negative sensor biases were considered, with two severity levels (moderate and severe faults) evaluated for each case.
	TSup_m1		-1 °K	
	TSup_p2		+2 °K	
	TSup_m2		-2 °K	

	TMix_p1	Mix air temperature sensor bias	+1 °K	Not all zone sensor faults were considered to avoid repetition. However, without the loss of generality, the south zone sensor faults for the cooling and shoulder season, and the east zone sensor faults for the heating season were considered.
	TMix_m1		-1 °K	
	TMix_p2		+2 °K	
	TMix_m2		-2 °K	
	TRet_p1	Return air temperature sensor bias	+1 °K	
	TRet_m1		-1 °K	
	TRet_p2		+2 °K	
	TRet_m2		-2 °K	
	RelPreAir_p15	Air loop differential pressure sensor bias	+15 Pa	
	RelPreAir_m15		-15 Pa	
	RelPreAir_p25		+25 Pa	
	RelPreAir_m25		-25 Pa	
	TAirSou_p1	South zone air temperature sensor bias	+1 °K	
	TAirSou_m1		-1 °K	
	TAirSou_p2		+2 °K	
	TAirSou_m2		-2 °K	
	TSupSou_p1	South zone discharging air temperature sensor bias	+1 °K	
	TSupSou_m1		-1 °K	
	TSupSou_p2		+2 °K	
	TSupSou_m2		-2 °K	
	VSupSou_flow_p15	South zone air flow rate sensor scale error	+15%	
	VSupSou_flow_m15		-15%	
	VSupSou_flow_p30		+30%	
	VSupSou_flow_m30		-30%	
	SupDucLea_10	Air loop supply duct leakage	FIR=10%	Implementing the mathematical expression as: $\dot{m}_{EF} + \dot{m}_L = k\sqrt{\Delta p}$, where $\dot{m}_{EF}$ is the normal operation mass flow rate, $\dot{m}_L$ is the leakage mass flow rate, k is the friction factor, and Δp is the pressure drop.
	SupDucLea_20		FIR=20%	
Duct/ Pipe	AirDucFou_10	Air loop duct fouling	FIR=10%	Increasing the nominal resistance of the duct/pipe with 10% or 20% magnitude.
	AirDucFou_20		FIR=20%	
	CHWPipFou_10	Chilled water pipe fouling	FIR=10%	
	CHWPipFou_20		FIR=20%	
	CHWPipIso_102L25	Chilled water pipe isolation	20 mm	The insulation of the 25 m pipe was damaged and its insulation thickness for the two scenarios
	CHWPipIso_1005L25		5 mm	

				was reduced to 20 mm and 5 mm, respectively (while assuming a perfectly insulated pipe for the baseline case).
Damper/ Valve	CooTowValLea_001	Cooling tower valve leakage	$l = 0.001$	The leakage ratio (l) in the damper/valve model was perturbed (with the value of 0.001 and 0.01) to represent moderate and severe fault levels, as defined below: $l = \frac{\kappa v(y = 0)}{\kappa v(y = 1)}$ where $\kappa v(y = 0)$ and $\kappa v(y = 1)$ are the flow coefficient at the fully closed and fully open positions, respectively.
	CooTowValLea_01		$l = 0.01$	
	ChiConValLea_001	Cooling water isolation valve leakage	$l = 0.001$	
	ChiConValLea_01		$l = 0.01$	
	ChiEvaValLea_001	Chilled water isolation valve leakage	$l = 0.001$	
	ChiEvaValLea_01		$l = 0.01$	
	CooCoiValLea_001	Cooling coil valve leakage	$l = 0.001$	
	CooCoiValLea_01		$l = 0.01$	
	OADamLea_001	Outdoor air damper leakage	$l = 0.001$	
	OADamLea_01		$l = 0.01$	
	CooCoiValStuck_0	Cooling coil valve stuck position	0% (fully closed)	The output of the damper/valve was overwritten to a fixed value when the fault was injected.
	CooCoiValStuck_5		5%	
	CooCoiValStuck_15		15%	
	CooCoiValStuck_65		65%	
	CooCoiValStuck_100		100% (fully open)	
	OADamStuck_0	Outdoor air damper stuck position	0% (fully closed)	
	OADamStuck_5		5%	
	OADamStuck_15		15%	
	OADamStuck_65		65%	
	OADamStuck_100		100% (fully open)	
HVAC mechanical fault	CooCoiFou_10	Cooling coil fouling	FIR=10%	Reducing the nominal overall heat transfer coefficient (UA) of the AHU cooling coil/tower by 10%, 30%, or 50% from the baseline in three fault scenarios.
	CooCoiFou_30		FIR=30%	
	CooCoiFou_50		FIR=50%	
	CooTowFou_10	Cooling tower fouling	FIR=10%	
	CooTowFou_30		FIR=30%	
	CooTowFou_50		FIR=50%	

	ChiConFou_20	Chiller condenser fouling	FIR=20%	Overwriting the original calculated chiller power by the fault ratio multiplier (found from empirical chiller fault models), as shown below. $\frac{\dot{W}_F}{\dot{W}_{EF}} = a_1 + a_2 T_{evap,out} + a_3 T_{cond,in} + a_4 \dot{Q}_{evap} + a_5 T_{evap,out} \dot{Q}_{evap} + a_6 T_{cond,in} \dot{Q}_{evap} + a_7 \dot{Q}_{evap}^2,$ where $\dot{W}_{EF}$ and $\dot{W}_F$ are the chiller power consumption under the fault-free and faulty conditions. $a_1 - a_7$ are the regression coefficients for the empirical models (each chiller mechanical fault has its own sets of regression coefficients). $T_{evap,out}$ and $T_{cond,in}$ are the evaporator outlet water temperature and condenser inlet water temperature. $\dot{Q}_{evap}$ is the heat transfer rate in the chiller evaporator.
	ChiConFou_40		FIR=40%	
	ChiNonCon_25	Chiller non-condensable gas	FIR=25%	
	ChiNonCon_50		FIR=50%	
	ChiOveCha_20	Chiller overcharging	FIR=20%	
	ChiOveCha_40		FIR=40%	
	ChiExcOil_30	Chiller excess oil	FIR=30%	
	ChiExcOil_60		FIR=60%	
	PumChiCavitation	Chilled water pump cavitation	-	
	PumChiImpeller	Chilled water pump impeller fault	-	
	PumChiMotDeg_15	Chilled water pump motor degradation	15%	
	PumChiMotDeg_30		30%	
	PumCWCavitation	Cooling water pump cavitation	-	overwriting the fault-free pump curve by the faulty pump curves under different fault types.
	PumCWImpeller	Cooling water pump impeller fault	-	
	PumCWMotDeg_15	Cooling water pump motor degradation	15%	
	PumCWMotDeg_30		30%	
	FanAHUMotDeg_15	Fan motor degradation	15%	
	FanAHUMotDeg_30		30%	
Control	ConkTSup_5	Supply air temperature PID loop fault	P=5	The proportional gains (P) of the Proportional-Integral-Derivative (PID) controller were perturbed to different order of magnitudes compared to the baseline setting. For instance,
	ConkTSup_05		P=0.05	
	ConkFanSpe_10	Supply fan speed PID loop fault	P=10	
	ConkFanSpe_01		P=0.01	

	ConkMinOut_5	Economizer PID loop fault	P=0.5	the normal P value for the cooling supply air temperature loop was 0.5. Accordingly, two fault scenarios with P value equal to 5 and 0.05 respectively were designed to represent the aggressive tuning and the slow response case.
	ConkMinOut_005		P=0.005	
	ConkPumSpe_1	Chilled water pump speed PID loop fault	P=1	
	ConkPumSpe_01		P=0.01	
	ConkCooTow_1	Cooling tower speed PID loop fault	P=1	
	ConkCooTow_01		P=0.01	
	ConVAVSoukCoo_1	South zone air cooling PID loop fault	P=1	
	ConVAVSoukCoo_01		P=0.01	
	ConVAVSoukDam_5	South zone VAV damper PID loop fault	P=5	
	ConVAVSoukDam_05		P=0.05	
	MaxCHWDifPreSP	Chilled water differential pressure setpoint falsely set to the maximum	-	-
	MaxCHWTSupSP	Chilled water supply temperature setpoint falsely set to the maximum	-	
	yCooDel_1	Cooling coil control signal delay	1 sec	Assuming the delay time of 1 s for certain communication protocol while it was assumed that the baseline case had a perfect delay-free communication.
	SupFanSpeDel_1	Supply fan speed control signal delay		
	yPumDel_1	Chilled water pump speed control signal delay		
Scheduling	Schedule_early_1	Schedule fault - Early termination of setback	1 hr	Adjusting the occupancy schedule parameters of the model.
	Schedule_early_2		2 hr	
	Schedule_delay_1	Schedule fault - Delayed onset of setback	1 hr	
	Schedule_delay_2		2 hr	
	Schedule_nosetback	Schedule fault - No temperature setback	-	
Design	Infiltration_70	More building infiltration than normal (poor airtightness design)	dp_nom = 70%	Overwriting the nominal differential pressure (dp_nom) between the building zone and the outdoor air to a smaller value under the same nominal mass flow rate, which indicates a decrease of the infiltration resistance.
	Infiltration_50		dp_nom = 50%	
	AHUFan_oversized	Oversized AHU supply fan	Next sizing in the catalogue	Changing the nominal values such as the nominal flow rate, nominal differential pressure, or the nominal performance curves of HVAC equipment.
	AHUCooCoil_oversized	Oversized AHU cooling coil		
	CHWPump_oversized	Oversized chilled water pump		
	CWPump_oversized	Oversized cooling water pump		
	Chiller_oversized	Oversized chiller		

Table 3. Details of *G36-1wk* fault scenarios in the heating season. Faults were injected continuously for a total of one week, but the system operated in occupied mode only from 7 a.m. to 8 p.m. each day.

Category	Fault abbreviation (CSV file name)	Fault type	Fault intensity	Method of fault imposition
Sensor	THWSup_p1	Hot water supply temperature sensor bias	+1 °K	Refer to the 'Sensor' faults explanation in Table 2.
	THWSup_m1		-1 °K	
	THWSup_p2		+2 °K	
	THWSup_m2		-2 °K	
	RelPreHW_p5000	Hot water differential pressure sensor bias	+5,000 Pa	
	RelPreHW_m5000		-5,000 Pa	
	RelPreHW_p10000		+10,000 Pa	
	RelPreHW_m10000		-10,000 Pa	
	Vout_p15	Outdoor air flow rate sensor scale error	+15%	
	Vout_m15		-15%	
	Vout_p30		+30%	
	Vout_m30		-30%	
	TSup_p1	AHU supply air temperature sensor bias	+1 °K	
	TSup_m1		-1 °K	
	TSup_p2		+2 °K	
	TSup_m2		-2 °K	
	TMix_p1	Mix air temperature sensor bias	+1 °K	
	TMix_m1		-1 °K	
	TMix_p2		+2 °K	
	TMix_m2		-2 °K	
	TRet_p1	Return air temperature sensor bias	+1 °K	
	TRet_m1		-1 °K	
	TRet_p2		+2 °K	
	TRet_m2		-2 °K	
	RelPreAir_p15	Air loop differential pressure sensor bias	+15 Pa	
	RelPreAir_m15		-15 Pa	
	RelPreAir_p25		+25 Pa	

	RelPreAir_m25		-25 Pa	
	TAirEas_p1	East zone air temperature sensor bias	+1 °K	
	TAirEas_m1		-1 °K	
	TAirEas_p2		+2 °K	
	TAirEas_m2		-2 °K	
	TSupEas_p1	East zone discharging air temperature sensor bias	+1 °K	
	TSupEas_m1		-1 °K	
	TSupEas_p2		+2 °K	
	TSupEas_m2		-2 °K	
	VSUPeas_flow_p15	East zone air flow rate sensor scale error	+15%	
	VSUPeas_flow_m15		-15%	
	VSUPeas_flow_p30		+30%	
	VSUPeas_flow_m30		-30%	
Duct / Pipe	SupDucLea_10	Air loop supply duct leakage	FIR=10%	Refer to the 'Duct/Pipe' faults explanation in Table 2.
	SupDucLea_20		FIR=20%	
	AirDucFou_10	Air loop duct fouling	FIR=10%	
	AirDucFou_20		FIR=20%	
	HWPipFou_10	Hot water pipe fouling	FIR=10%	
	HWPipFou_20		FIR=20%	
	HWPipIso_102L25	Hot water pipe poor isolation	20 mm	
	HWPipIso_1005L25		5 mm	
Damper/ Valve	BoiValLea_001	Boiler isolation valve leakage	$l = 0.001$	Refer to the 'Damper/Valve' faults explanation in Table 2.
	BoiValLea_001		$l = 0.01$	
	BoiBypValLea_001	Boiler bypass valve leakage	$l = 0.001$	
	BoiBypValLea_01		$l = 0.01$	
	HeaCoiValLea_001	Heating coil valve leakage	$l = 0.001$	
	HeaCoiValLea_01		$l = 0.01$	
	OADamLea_001	Outdoor air damper leakage	$l = 0.001$	
	OADamLea_01		$l = 0.01$	
	HeaCoiValStuck_0	Heating coil valve stuck position	0% (fully closed)	
	HeaCoiValStuck_5		5%	
	HeaCoiValStuck_15		15%	
	HeaCoiValStuck_65		65%	
	HeaCoiValStuck_100		100% (fully open)	

	OADamStuck_0	Outdoor air damper stuck position	0% (fully closed)	
	OADamStuck_5		5%	
	OADamStuck_15		15%	
	OADamStuck_65		65%	
	OADamStuck_100		100% (fully open)	
HVAC mechanical fault	HeaCoiFou_10	heating coil fouling	FIR=10%	Reducing the nominal overall heat transfer coefficient (UA) of the AHU heating coil by 10%, 30%, or 50% from the baseline in three fault scenarios.
	HeaCoiFou_30		FIR=30%	
	HeaCoiFou_50		FIR=50%	
	BoiFou_20	Boiler fouling	FIR=20%	Adjusting down the boiler efficiency curve (by 10% and 20%) since deposits on the fireside and the waterside of the boiler tubes could impair the heat transfer and reduce the boiler efficiency.
	BoiFou_40		FIR=40%	
	PumHWCavitation	Hot water pump cavitation	-	Overwriting the fault-free pump curve by the faulty pump curves under different fault types.
	PumHWImpeller	Hot water pump impeller fault	-	
	PumHWMotDeg_15	Hot water pump motor degradation	15%	
	PumHWMotDeg_30		30%	
	FanAHUMotDeg_15	Fan motor degradation	15%	
	FanAHUMotDeg_30		30%	
Control	ConkTSup_5	Supply air temperature PID loop fault	P=1	Refer to the 'Control' faults explanation in Table 2.
	ConkTSup_05		P=0.01	
	ConkFanSpe_10	Supply fan speed PID loop fault	P=10	
	ConkFanSpe_01		P=0.01	
	ConkMinOut_1	Economizer PID loop fault	P=0.1	
	ConkMinOut_001		P=0.001	
	ConkPumHWSpe_5	Chilled water pump speed PID loop fault	P=5	
	ConkPumHWSpe_05		P=0.05	
	ConVAVSoukCoo_1	South zone air cooling PID loop fault	P=1	
	ConVAVSoukCoo_01		P=0.01	
	ConVAVSoukDam_5		P=5	

	ConVAVSoukDam_05	South zone VAV damper PID loop fault	P=0.05	
	MaxHWDifPreSP	Hot water differential pressure setpoint falsely set to the maximum	-	
	MinHWTSupSP	Hot water supply temperature setpoint falsely set to the minimum	-	
	yHeaDel_1	Heating coil control signal delay	1 <i>sec</i>	
	SupFanSpeDel_1	Supply fan speed control signal delay		
	yPumHWDel_1	Hot water pump speed control signal delay		
Scheduling	Schedule_early_1	Schedule fault - Early termination of setback	1 <i>hr</i>	Refer to the 'Scheduling' faults explanation in Table 2.
	Schedule_early_2		2 <i>hr</i>	
	Schedule_delay_1	Schedule fault - Delayed onset of setback	1 <i>hr</i>	
	Schedule_delay_2		2 <i>hr</i>	
	Ichedule_nosetback	Schedule fault - No temperature setback	-	
Design	Infiltration_70	More building infiltration (poor airtightness design)	dp_nom = 70%	Refer to the 'Design' faults explanation in Table.
	Infiltration_50		dp_nom = 50%	
	AHUFan_oversized	Oversized AHU supply fan	Next equipment sizing in the catalogue	
	AHUHeaCoil_oversized	Oversized AHU heating coil		
	HWPump_oversized	Oversized hot water pump		
	Boiler_oversized	Oversized boiler		

Additional information

- Additional helpful document(s):
 - Lu, X., Fu, Y., O'Neill, Z. & Wen, J. A holistic fault impact analysis of the high-performance sequences of operation for HVAC systems: Modelica-based case study in a medium-office building. *Energy and Buildings* **252**, 111448 (2021).
- No specific holidays were considered unoccupied days in this dataset.

- Please note, since this dataset represents a simulated building, the baseline (fault-free) data was intentionally generated with the same timestamp as the fault data. This was achieved by applying identical outdoor conditions (e.g., outdoor air humidity and temperature), occupancy load, time of day, simulation period, and other related parameters when generating the dataset from the simulated model. This approach allows researchers to compare the corresponding baseline and fault/attack datasets without confounding variations from external factors.
- Fig. 1 shows the schematic of the corresponding AHU system. The system is a single-duct multi-zone VAV AHU connected with five VAV terminal boxes that serve five zones (four exterior and one interior zone) on a single floor.

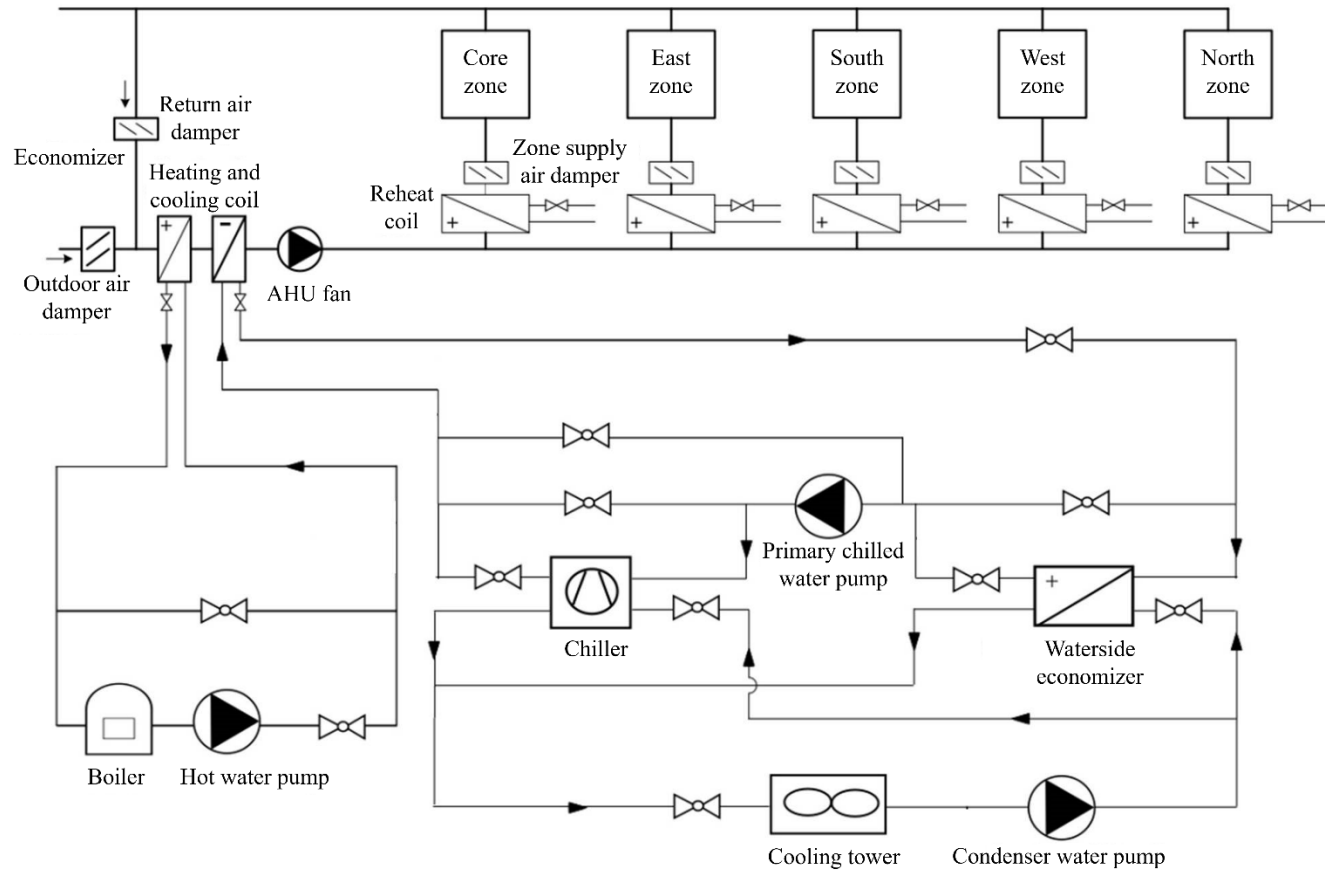


Fig. 1. Schematic of the AHU diagram, used to generate the dataset.

- As described in the original document, the simulated model (modeled in Modelica, Dymola) is based on and verified against the medium-sized office prototype model developed by the U.S. Pacific Northwest National Lab in EnergyPlus. The original EnergyPlus model has three floors, but only the middle floor, featuring 5 zones with a variable air volume (VAV) air handling unit (AHU), was simulated here for simplicity. Additionally, a chilled water plant system and a boiler hot water system were added to the model. The detailed sizing parameters and the comparison to the original EnergyPlus prototype model are included in Table 4. Both air-side and plant-side high-performance operation sequences implemented in the virtual testbed are also summarized in Table 5.

Table 4. Comparison between the EnergyPlus prototype with the Dymola model

Item	EnergyPlus midfloor plenum	Dymola model
Area [m ²]	1660.7	1662.7
AHU fan design flow rate [m ³ /s]	4.2	4.8
AHU fan head [Pa]	1389	1381
Overall efficiency	0.6	0.49
AHU fan rated electric power [W]	9685	13530
Cooling coil capacity [W]	95438	100711
Heating coil capacity [W]	34995	40526
Chilled water pump head [W]	Not applicable	255000
Chilled water pump flow [m ³ /s]		0.004
Cooling water pump head [W]		215700
Cooling water pump flow [m ³ /s]		0.0043
Hot water pump head [W]		157000
Hot water pump flow [m ³ /s]		0.00132
Cooling tower fan power [W]		4300

Table 5. Summary of the implemented control sequences in both air-side and plant-side.

Category	Equipment	Control sequence
Air side	Air handling unit	Fan speed control Economizer damper control Minimum outdoor air control Supply air temperature control
	Zone VAV terminal	Zone air temperature control VAV box airflow and reheat coil control
Plant side	Supervisory	AHU and zone mode control

	Chiller	Chiller staging control Minimum flow bypass valve control Chilled water supply temperature reset control
	Cooling tower	Cooling tower fan speed
	Pump	Chilled water pump speed control Hot water pump speed control
	Boiler	Boiler staging and heating power control Minimum flow bypass valve control Hot water supply temperature reset control
	Supervisory	Chiller/boiler plant enable/disable control Cooling mode control (free cooling, partial mechanical cooling, full mechanical cooling)